



Technische Hochschule
Ingolstadt

Trendbook

EXPO 2035

Political / Legal

Digital Age Governance

Evolving Laws for AI Usage, Data Privacy and Digital Governance



Key uncertainties:

- Guided by policies such as the GDPR, a worldwide united digital landscape develops. AI is transparent, citizens have ownership over their data, and digital governance fosters democracy, equity, and trust.
- Lack of global alignment leads to unchecked data use. While some regions regulate, others prioritize control or profit. Surveillance grows, trust erodes, and citizens lose digital rights.

Description:

- AI is rapidly leading decision-making in urban planning, healthcare, law, and business.
- The speed and complexity of AI are outpacing traditional data privacy laws and governance frameworks.
- Autonomous, adaptable AI systems raise new moral and legal concerns about responsibility, transparency, and human rights
- Governments and global organizations face pressure to create adaptable, future-proof legal systems.
- To ensure technology benefits society fairly and securely, it is essential to focus on global cooperation, adaptable laws, strong digital rights, effective digital governance, and data privacy protection.
- Rising push to create legal rules for digital identity systems.

Relevance / Impact:

- Public demand for algorithmic transparency and ethical oversight will increase.
- New global alliances could form around shared digital governance principles.
- Businesses face increased costs and complexity in navigating diverse regulatory landscapes.

Weak Signals (Proofs /Examples):

- Rise in privacy-first design tools and AI compliance startups (e.g. Trustible, Ketch, Apheris)
- The EU AI Act and U.S. AI Order establish safeguards for AI safety, rights, and high-risk systems.
- Formation of cross-border digital governance bodies.
- Estonia pioneers' digital governance with blockchain technology, ensuring secure and transparent public services.



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Rise of Sustainability-Driven Legislation



Key uncertainties: - Global actors: governments, businesses, and the public either coordinate to champion unified sustainability laws or create friction through resistance and fragmented regulations. - Legal systems could become powerful enforcers of climate justice or be weakened by poor enforcement, politicization, judicial conservatism, and growing resistance from industries and the public.	
Description: - Governments worldwide realize that voluntary efforts are not enough to tackle the climate crisis. - They are introducing legally binding sustainability laws to cut carbon emissions, preserve resources, and promote circular economies. - These regulations now influence trade policy, financial regulation, urban planning, and corporate governance, not just environmental ministries. - As a result, industries must innovate, adopt cleaner energy, build greener supply chains, and design eco-friendly products.	
Relevance / Impact: - Showcase progress in sustainable legislation and green policy implementation. - Exchange solutions on how legal frameworks can drive ecological innovation and climate resilience. - Discuss harmonization of sustainability laws across borders to facilitate responsible globalization.	Weak Signals (Proofs /Examples): - Regulatory shift toward mandatory sustainability disclosures. - Legal accountability for corporate emissions rising. - Trade Policy being used to enforce Climate standards. - Developing nations adopting strict environmental laws. - Climate data becoming a regulatory requirement



Key sources: [UN Environment Programme \(UNEP\)](#) , [World Economic Forum \(WEF\) – Global Risks Report](#) , [UN- SDGs](#)

The Rise of Exclusion: Nationalism, Xenophobia, and Social Division



Key uncertainties:

- Increased Migration flows and demographic shifts could lead to an increase in national identities.
- Unsuccessful implementation of online content regulations for misinformation and hate speech could lead to increased social division and spreading of nationalist propaganda.

Description:

- Intensified emphasis on national identity and cultural protectionism.
- Prevalence of xenophobic and racist narratives creating less inclusive environments.
- Increased social divisions within and between nations.
- Exploitation of nationalist sentiments by political leaders.
- Increased border controls and travel restrictions.

Relevance / Impact:

- Increased security risks and potential for politically motivated disruptions.
- Challenges to maintaining inclusivity and fostering genuine cultural exchange.
- Risk of events being exploited for nationalistic messaging and political agendas

Weak Signals (Proofs /Examples):

- Increasing electoral success of populist and nationalist parties globally (e.g., Brexit, Trump).
- Growing use of xenophobic and ethnocentric rhetoric in public discourse.
- Implementation of protectionist economic.
- Increasing reports of hate crimes targeting minority groups and immigrants.



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Multi-Polarization

Navigating a World Without a Single Power Center



Key uncertainties:

- Will the world move toward more cooperation among regional powers or face growing competition and division?
- As countries like China, India, Brazil, and Russia become stronger, will they build stability or act in their own interest?
- Can old global institutions like the UN, IMF, and WTO keep up with change, or will new systems replace them?
- Will there be common rules for tech like AI and data, or will each region follow its own standards?

Description:

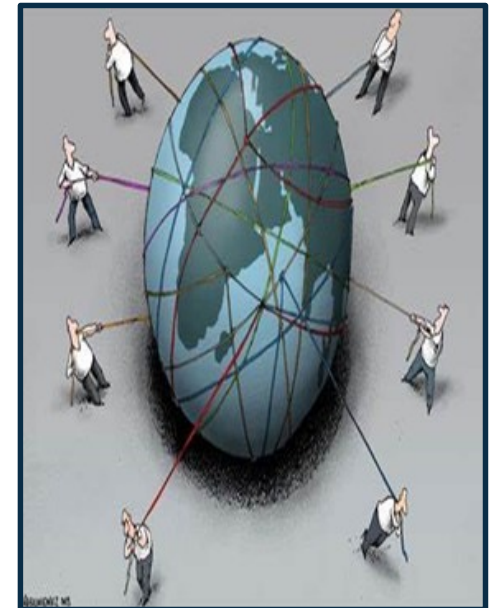
- Global influence is no longer concentrated in a few hands. Instead, power is spread across different regions.
- More countries are following their own paths in development, defense, and governance instead of copying western models.
- Different political and economic styles like democracy, state-led capitalism, and digital authoritarianism exist and compete at the same time.
- Instead of joining long-term alliances, many countries now work together on specific issues like climate, trade, or tech when it benefits them.

Relevance / Impact:

- Emerging Economies like India and China, now have a bigger role in shaping the global agenda.
- Solving global problems like climate change or cyber threats is harder because countries don't always agree.
- It's more difficult to do business across borders due to mismatched trade rules, tech standards, and investment laws.
- Many countries are trying to reduce dependence on global powers by becoming more self-sufficient in areas like energy, tech, and defense.

Weak Signals (Proofs /Examples):

- The inclusion of new members and discussion of a BRICS(Inter-Gov. organization consisting of 10 Emerging Economies) currency shows efforts to move away from dollar dominance. (1) (4)
- The AU's new G20(Inter-Gov. Organization of 19 countries and EU) and seat represents growing recognition of Africa's voice in global decisions. (2)
- India is sharing platforms like UPI and digital ID systems with other countries.
- Both EU and ASEAN are pushing for regional control over tech infrastructure and rules. (3)
- Countries are shaping their own AI rules, creating global divergence in ethics and standards.



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Key sources: 1. [BRICS Nation on replacing USD](#) ; 2. [African Union Joins G20](#) ; 3. [EU and ASEAN Push for Tech Sovereignty](#) ; 4. [China ramps Yuan push](#)

Influence of Technology on Politics

Emerging Frontiers of Technology-Driven Governance and Civic Engagement



Key uncertainties:

- Can societies effectively counter AI-driven deception?
- Can democracies navigate hyperconnected info landscapes, will the digital tools bridge divides or deepen polarization?
- Will digital voting systems earn public confidence?
- Could decentralized tech reshape how we vote and govern?

Description:

- Technology is transforming political discourse, decision-making, and civic engagement enables more responsive and inclusive governance through digital tools.
- Hyperconnectivity fuels both grassroots mobilization and political polarization.
- AI drives voter microtargeting via sentiment analysis and behavioral insights raises ethical concerns around privacy, manipulation, and algorithmic bias.
- E-voting and e-citizenship enhance accessibility, especially for remote populations.
- Cyber warfare by state and non-state actors disrupts political stability.
- Demand for robust cybersecurity and trustworthy digital identity systems is rising.
- Political digitalization requires agile, forward-looking regulation to protect democratic integrity.

Relevance / Impact:

- Digital Civic Participation expands political activism and communication; raises manipulation and influence risks.
- Increased vulnerability to digital deception and cyberattacks.
- Need for agile, tech-savvy policymaking.
- Cyber Warfare undermining election integrity and national stability.
- Cryptocurrency disrupts traditional monetary systems

Weak Signals (Proofs /Examples):

- 5G/6G scaling civic tech infrastructure thus enhancing the hyperconnectivity
- Blockchain voting pilots in Estonia and Switzerland
- Deepfakes spreading disinformation and used in election campaign.
- AI tools in civic education platforms
- Cyberattacks targeting electoral systems



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Technological Expansion of Government Control

The Rise of Tech-Driven State Control: Surveillance, Censorship, and Digital Governance



Key uncertainties:

- Citizens resist surveillance and censorship, governments are pressured to implement digital governance that protects privacy, civil liberties, and human rights.
- People accept control and global norms fragment, authoritarian states normalize surveillance, censorship, and centralized digital systems without external checks.

Description:

- Governments are using technologies like surveillance systems, AI, and biometrics to monitor and control citizens' activities
- Digital currencies, e-government platforms, smart cities provide centralized control over finances, services, and urban life
- Internet censorship and social media manipulation are being employed to restrict access to foreign information and shape public opinion.

Relevance / Impact:

- Increased surveillance and monitoring of citizens' activities
- Enhanced efficiency in governance
- Centralization of power in government control
- Erosion of civil liberties, privacy, freedom of speech
- Geopolitical tensions arise as nations adopt conflicting digital governance models and control measures
- Stifling opposition with internet censorship and government-run propaganda

Weak Signals (Proofs /Examples):

- Increased deployment of facial recognition or biometric systems in non-government sectors [1]
- The pilot programs for "smart cities" (China, USA, Canada) with extensive monitoring [2]
- pilot programs for e-government services and digital currencies [3]
- Increased Use of "Firewall" Technologies and censorship by Governments [4]



Key sources: [1. UK Briefing on Facial Recognition](#); [2. China's Smart Cities Development](#), [USA](#), [Canada](#); [3. Digital currencies](#), [E-Estonia](#); [4. China's Firewall](#), [Russian Internet Control](#), [Ban media in Germany](#)

Economic / Environmental

Carbon Neutrality by 2045: Replacing Coal with Hydrogen in the Production of Steel



Key uncertainties:

- High production cost of clean hydrogen vs fossil fuels
- Risk of facing pressure from countries that produce steel using coal

Description:

Forging a cleaner future:

Replacing coal with **clean hydrogen** in the production of steel marks a transformative step towards a climate-neutral industry. Steelmaking typically relies heavily on coal. This traditional method to produce steel emits substantial amounts of CO₂—accounting for approximately 7% of Germany’s total CO₂ emissions [1]. Given that a significant portion of the infrastructure for **Berlin’s EXPO** is expected to be made of steel, this presents a potential challenge to Germany’s carbon neutrality ambitions. To address this issue, the integration of clean hydrogen into the steelmaking process offers a promising solution. Unlike coal, which releases CO₂ when used in blast furnaces to reduce iron ore into molten iron, hydrogen reduces iron ore to iron sponge, producing only **water vapor** (H₂O) as a byproduct [2].



Relevance / Impact:

- The technological shift has the potential to nearly eliminate carbon emissions from Germany’s steel sector.
- Compliance with EU carbon regulations
- Replacing coal in heavy industry will aid Germany’s in its quest to transition its entire energy system to renewables.
- Embracing hydrogen secures Germany’s role as a global leader in sustainable industrial innovation, thus avoiding falling behind in green tech competitiveness.

Weak Signals (Proofs /Examples):

- Rising Investor Interest in Low-Carbon Steel [3]
- Emergence of Green Hydrogen Startups and Ecosystems [4]
- Germany’s new infrastructure plans to transport hydrogen efficiently from production sites to industrial consumers [5].

Key sources:

- [1]- Cara Bien (2023). *Wasserstoff statt Kohle: Die Stahlproduktion in Deutschland wird grün*
- [2]- Sander-Green, Lars (2024). *Do We Really Need Coal to Make Steel?*
- [3]- Payton Ben (2023). *Steelmakers Face Investor Pressure to Accelerate Transition.*
- [4]-Frede et al. (2024). *Mind the Gap: Venture Funding of Hydrogen Start-ups*
- [5]- Collins, Jennifer (2025). *Construction on Germany’s Hydrogen Core Grid to Commence in 2025.*

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Carbon Neutrality by 2045: Local Production of Energy Through Cooperatives



Key uncertainties:

- Will Berliners embrace local energy cooperatives as a step toward a sustainable future, or will they perceive them as a barrier to convenient energy access?
- Will innovations such as smart grids and instant trading platforms scale efficiently and securely to enable managing real-time energy trading between thousands of users?

Description:

Energy by and for the people:

Local energy cooperatives play a vital role in Berlin's journey towards carbon neutrality. This is done through empowering communities to generate, manage, and consume renewable energy at a decentralized level [1]. These citizen-led initiatives, such as **BürgerEnergie Berlin** [2], support the installation of solar panels on municipal and residential buildings, promote democratic participation in the energy transition, and reduce dependence on fossil fuels. These cooperatives can serve as a compelling showcase for **Berlin's EXPO**, highlighting innovation and sustainability by demonstrating how cities can decentralize power generation while strengthening energy security and encouraging youth engagement in building a carbon-free future [3].

Relevance / Impact:

- Reduced reliance on fossil fuel energy plants.
- Cooperatives often prioritize the production of clean energy.
- Residents benefit from cooperatives via lower energy costs.
- Energy cooperatives often lead to innovative methods such as peer-to-peer energy trading via blockchain

Weak Signals (Proofs /Examples):

- Cooperatives in Berlin are exploring blockchain tech. for direct energy trading among citizens [4].
- Younger demographic within Germany are showing increased interest in funding energy co-ops. [5]
- Berlin senate proposes initiative to focus on locally generated energy [6]



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Key sources:

[1] Hua Depeng, Francisco Javier S. Lacarcel & Virginia Simón-Moya (2024). *The dynamics of energy communities and innovative cooperatives: Mapping current knowledge and future trends*

[2]- BürgerEnergie Berlin – Beteiligung an der Berliner Energieinfrastruktur!

[3]- Oscar Güell (2023). *Social Impact of energy communities and the Benefits they Bring*

[4]-Chloe Jardine (2021). *Germany Launches Blockchain Energy Trading Project*

[5]- Suzanne Goldenberg (2013). *Berliners' co-op aims to take over and run electricity grid of city*

[6]-Green hydrogen: Berlin introduces Federal Council initiative (2024). Available from: [Green hydrogen: Berlin introduces Federal Council initiative – Berlin.de](https://www.berlin.de/en/berlin/energy-and-climate/green-hydrogen/berlin-introduces-federal-council-initiative)

Transforming the Financing of Mass Events by 2035

Innovative funding models to promote inclusivity and growth through Public-Private Partnership.



Key uncertainties:

- Influence of corporates and big companies on mass events that can lead to targeted promotions.
- The event turning out to be profit making exercise rather than a solution-based exhibition.

Description:

Hosting a world mass event costs considerable amount of public spending in infrastructure and mobilization of people. Not only the host city but also different participating countries and/or organizations spend millions of dollars just to set up the pavilions, most of them are abandoned after the event at the current stage.

Dedicated Event Sustainability Funds pool can be set up to finance the events. Green Bonds, Decentralized Finance (DeFi) and Tokenized assets (Micro-Investments) may be used for investments in mass events.

The events can be jointly hosted by different nations with digital twins present at either cities of the countries to share costs.

The investment can be distributed among key players, i.e organizations and nations, to make the exercise more sustainable.

The infrastructure and exhibition sites can be used for commercial activities. Public amenities or corporate offices, that can be used to make profits on the condition of minimal influence.

Stock listings or public-private consortiums can be set up where corporations or visitors can hold equity stakes.

Relevance / Impact:

- The cost benefits will make it more inclusive for general public and developing nations.
- A revenue sharing model can be designed to attract investors.
- The concept of profits will encourage investors to innovate and make sustainable infrastructure to be used for commercial purposes after the event.
- The future of mass events is not just the exhibition for a few months or years, but an entrepreneurial activity that feeds off the investment to make a sustainable business out of it.

Weak Signals (Proofs /Examples):

- Osaka (Expo 2025) issued ¥5 billion in green bonds to fund Expo infrastructure, linking it explicitly to SDG goals
- US has been financing its pavilion by private funding.

Key sources:

1. <https://www.nomuraholdings.com/news/nr/nsc/20240119/20240119.html#:~:text=Tokyo%2C%20January%2019%2C%202024%E2%80%9494Nomura%20Securities,va%20at%20five%20billion%20yen>
2. <https://www.inparkmagazine.com/the-u-s-should-fund-its-own-expo-pavilions/>
3. <https://www.weforum.org/stories/2024/12/tokenization-blockchain-assets-finance/>
4. https://www.investkorea.org/ik-en/bbs/i-5004/detail.do?ntt_sn=490779#:~:text=2.%20A%20Carbon

Transforming the Financing of Mass Events by 2035



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Circular Economy at EXPO 2035

Designing Sustainable Exhibitions: A Living Lab for Circular Innovation and Sustainable Industry



Key uncertainties:

- Strengthening of global event sustainability standards by 2035 (mandatory requirements)
- Reduction of waste and growth of material recycling in the global economy.

Description:

In Germany, buildings account for around 35 % of Germany's energy consumption and around one third of its CO₂ emissions. Therefore, we need to **fundamentally rethink the way we design, produce and operate our built environment; but above all, how we post-purpose or repurpose it.** [1]

Expo 2035 will showcase the circular economy as a living industrial laboratory for sustainable modernization. The focus will be on **reuse, recycling, and rethinking of all resources** — materials, energy, and logistics — during the construction of pavilions, the infrastructure of new districts of the Berlin and the modernization of the current infrastructure of the Berlin - as a sustainable modern city of the future.

A Living Lab for Circular Innovation and Sustainable Industry will include modular, reusable pavilions and eco-design principles. It will show minimal waste production and second-life strategies for exhibits. It will also show the use of smart energy systems with low-consumption materials and present the integration of renewable energy sources with adaptive infrastructure. This lab will also be presented as a VR demonstration of circular practices in sectors like construction, energy, manufacturing, and urban infrastructure.

Formats: Physical (pavilions, real projects), Hybrid (laboratories with live demonstrations, interactive training sessions), Virtual (VR tours, online sessions, knowledge hub).

Relevance / Impact:

- Strong demand for environmentally responsible event management.
- Pressure from governments, UN, and ESG standards to minimize event waste.
- Higher attractiveness to sponsors focused on sustainability branding.
- Competitive advantage for countries and companies presenting circular solutions.

Weak Signals (Proofs /Examples):

- Use of fully recyclable materials in pavilions (EXPO 2020 Dubai, EXPO 2025 Osaka) [1].
- Advancing the Energy Transition: structure, aims and topics of the Kopernikus projects [2].
- The European Commission's 2050 Vision "A clean planet for all" (Final report). [3]
- Germany NZEB-Goals (nearly zero energy buildings) - Reduction in the primary energy requirement by 80% by 2050. [4]
- Implementation of "zero waste" policies, growth of modular architecture at global exhibitions and conferences [5].
- The European Commission adopted the new circular economy action plan (CEAP) in March 2020. [6]
- The plastics manufacturer Covestro pursues the vision of becoming fully circular and offers a continuously growing range of circular products. [7]

Key sources:

1. <https://expo2025germany.de/nachhaltigkeit-en/>
2. <https://www.kopernikus-projekte.de/en/projects>
3. https://www.umweltbundesamt.de/sites/default/files/medien/479/publikation/en/cc_17-2022_the_european_commissions_2050_vision.pdf
4. <https://nzeb.in/>

5. <https://op.europa.eu/en/publication-detail/-/publication/92f6d5bc-76bc-11e9-9f05-01aa75ed71a1>
6. https://environment.ec.europa.eu/strategy/circular-economy-action-plan_en
7. https://www.deloitte.com/de/de/Industries/energy-chemicals/case-studies/innovativer-circular-economy-vermarktungsansatz.html?id=de:2ps:3gl:4__ec_story-covestro:5:6er:20250402::sm&gad_source=1&gad_campaignid=22406024041&gbraid=0AAAAAD-dxRIItA34jd6vPbHc-2Jbglcg03&gclid=Cj0KCQjw2tHABhCiARIsANZzDWqefEHQvp-jetZwPJJAEO8G-IR0xbArcJ174Y0Ja11DvxHx8FBrkYaAkPMEALw_wcB



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AgroFuture: Biotechnology for the Planet and Food Security

Sustainable Bio-Economy: “City smart farming, Sustaining the planet”



Key uncertainties:

- Extreme weather volatility and climate adaptation
- Will the production of food using biotechnology become concentrated in cities?
- Will public opinion shift toward accepting biotechnologies in food production?

Description:

- By 2035, farms will become intelligent, dynamic ecosystems that automatically cope with growing crop, soil, and weather conditions.
- Climate-Responsive Smart Farming Ecosystem systems optimize yields in real time while maintaining resources and shielding crops from climate volatility through the implementation of AI, robotics, and bio-adaptive materials.
- By developing vacant buildings, rooftops, and vertical areas into beneficial agricultural sectors, biotechnological agriculture is transforming the urban landscape.
- Localised food production increases circular economies and urban employment while reducing transportation emissions and logistics expenses.
- Innovations support Europe's transition to more sustainable farming practices by lowering greenhouse gas emissions, combating land degradation, conserving water, and reducing chemical pollution in ecosystems.

Relevance / Impact:

- Stronger food security under climate change conditions.
- Strengthening Europe's leadership in agricultural biotechnology.
- Urban agriculture promotes community cohesion and food literacy by reviving people's connection to the source of their food.

Weak Signals (Proofs /Examples):

- AeroFarms is a large-scale vertical farm utilizing resource-efficient indoor farming that could be adopted in urban settings.
- Local Bounti operates advanced indoor farms using controlled environment agriculture to grow leafy greens and herbs.
- Lufa Farms built the world's first commercial rooftop greenhouse and now has multiple facilities.
- Brooklyn Grange serves several large-scale commercial rooftop farms, growing a variety of vegetables and flowers.

Key sources:

1. <https://www.aerofarms.com/>,
2. <https://localbounti.com/>
3. <https://montreal.lufa.com/en>,
4. <https://www.brooklyngrangefarm.com/>



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Preparedness for Climate Change through Sustainable Urban Development



Key uncertainties:

- Change in policy due to change in government could lead to slow or reverse the Sustainable Urban Development progress.
- The under estimated rate of climate change could outpace climate resilient infrastructure adaptation.
- The resistance by public due to lack of awareness for the adoption of required policy changes.

Description:

Berlin as a global model city demonstrating how a city can be resilient to climate change.

Water Management Systems: Combat urban flooding and improve groundwater recharge by the use of Permeable Pavements, Urban Flood Buffers, Smart Water Grids, Rain Gardens.

Urban Heat Mitigation: Reduce heat stress and heat island effect by the use of Shade Corridors, Misting Systems, Urban lakes.

Green Roofs and Urban Forests: Tackle heat islands, improve air quality and to promote urban biodiversity by the use of Green Roof Mandates, Vertical Forests, Micro-forests.

Emergency Preparedness: Enhance the resilience to climate extremes by the use of Climate Proof Infrastructure, Cooling Shelters, Drought Combat Plan, Digital Twin Model.

Relevance / Impact:

- **Local Impact:** Reduces risk to Berlin's critical infrastructure, enhances public health, and boosts economic resilience.
- **Global Relevance:** Berlin can act as a climate urban lab, influencing policy and design in other global cities.
- **Expo2035 Legacy:** Permanent climate-resilient installations, educational outreach, and innovation hubs promoting sustainable living.

Weak Signals (Proofs /Examples):

- **Urban Climate Insurance:** Cities exploring municipal-level climate insurance products.
- **AI-Powered Green Planning:** Use of AI in microclimate mapping for better placement of green and blue infrastructure.
- **Climate Migration:** Small but growing populations relocating due to heat or water stress may impact Berlin's urban dynamics.
- **Bioadaptive Architecture:** Buildings that respond in real time to environmental changes (shape-shifting facades, self-healing materials).

Key sources:

1. <https://www.umweltbundesamt.de/en/topics/climate-energy/climate-impacts-adaptation/adaptation-to-climate-change>
2. <https://unhabitat.org/knowledge/best-practices>
3. <https://www.bmu.de/en/download/german-strategy-for-adaptation-to-climate-change-summary>

4. <https://iclei.org/activity/2024-climate-risk-and-vulnerability-assessment-cohort/>
5. <https://www.c40.org/case-studies/>
6. <https://www.ipcc.ch/report/ar6/wg2/chapter/summary-for-policy-makers/>



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Socio-Cultural

Youth Culture : The Social Shift in Global Cities



Key uncertainties:

- Will cities invest in inclusive spaces for cultural expression (e.g., music venues, public art zones) ?
- If urban policies favor commercialization or gentrification, grassroots culture may be pushed out or diluted.
- Will local cultures continue to thrive or be homogenized by global trends?

Description:

- By 2050, 68% of the world’s population is projected to live in urban areas (UN). This shift increases the influence of urban youth as the primary shapers of global culture, politics, and economy. What once started as underground hip-hop, sneaker culture, or street fashion—has gone mainstream, largely due to youth leadership and authenticity. This shift has redefined modern urban identity worldwide. Global cities like New York, Lagos, Seoul, São Paulo, and Berlin are epicenters where young people create and remix cultural forms—music genres, fashion styles, and visual arts—turning local trends into global movements.
- Young leaders are poised to continue shaping culture, actively redefining the future of creativity in urban environments.

Relevance / Impact:

- Youth-led culture fuels the creative economy—music, gaming, fashion, and content creation. In cities, these sectors generate jobs and innovation, especially where traditional industries are declining.
- As the world urbanizes (with 68% of the population in cities by 2050), youth culture becomes a key lens through which to understand climate resilience, inequality, migration, and social cohesion in densely populated areas.
- From hip-hop in the Bronx to K-pop in Seoul, urban youth cultures often spark trends that become global phenomena.

Weak Signals (Proofs /Examples):

- Urban policies are progressively affirming creativity as a fundamental human right for all children, as exemplified by [Helsinki’s Culture Kids initiative](#) and comparable cultural programs in Paris.
- Young people are influencing cultural policy through platforms such as Dubai’s Youth Circles, [Hamburg’s Youth Manifesto](#) and [Hong Kong’s Youth Advisory Committee](#)
- Despite budget limits, cities like Abu Dhabi, Barcelona, Oslo, and Taipei are finding creative ways to expand arts education in schools.



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Key sources: [Dubai Youth Circle](#) , [Brazilian Creative Learning Network](#) , [2050 - World Urban Population Estimation](#), [Cool Minds Youth Advisory Committee](#)

24-Hour Cities Shaping the Socio-Cultural Future



Key uncertainties:

- How can cities balance a vibrant night economy with residents' quality of life?
- Are there protections in place for fair wages, work hours, and rest in the night economy?
- Continuous energy use (lighting, transport, cooling systems) has environmental costs.

Description:

- As public interest in inclusive and vibrant urban life grows, there is increasing demand for 24-hour cities that foster cultural exchange and social connectivity. Queer cultural spaces like Club Raum are being recognized not only as vital to a city's cultural infrastructure but also as key contributors to creative economies and social equity. A 24-hour city stays active and accessible around the clock, offering continuous public transportation, open cultural venues, late-night businesses, and essential public services.
- This evolving urban model responds to a growing call for spaces that support dynamic cultural and social life beyond traditional hours, and such cities are expected to become more common in the near future.

Relevance / Impact:

- 24-hour cafés, night markets, libraries, and community centers serve as meeting points for people of different backgrounds, fostering social interaction and cultural exchange.
- The concept of a 24-hour city is increasingly being shaped around the principle of safety and inclusivity, with particular emphasis on ensuring women's safety in public spaces at all hours.

Weak Signals (Proofs /Examples):

- A broader perspective on nightlife: Night culture extends beyond clubs, encompassing experiences like [Seoul's night markets](#) Dubai's Ramadan Nights, and [Buenos Aires' La Noche de las Librerías](#).
- Hamburg is actively pursuing initiatives to evolve into a 24-hour city, emphasizing enhanced mobility, inclusive cultural spaces, and a vibrant nighttime economy.



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Key sources: [Stockholm - A 24 Hour City](#), [China's Nighttime Economy](#), [Hamburg's Journey to a 24-Hour City](#), [Dubai's Ramadan Nights](#)

Identity Expression & Individualism



Key uncertainties:

- Will increasing individual expression lead to tension with traditional or conservative cultural norms at large-scale public events?
- Can mass events maintain inclusive values without tokenizing or commodifying identity?
- How will event design balance identity freedom with community cohesion and shared experiences?

Description:

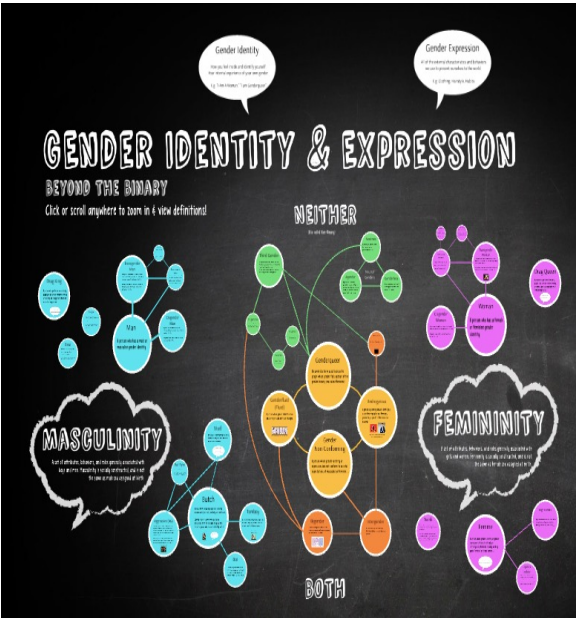
- People increasingly seek platforms—especially in mass events—to express and celebrate their personal, cultural, neurodiverse, and gender identities. What was once niche is now mainstream: inclusion, authenticity, and individuality have become central design pillars in events, festivals, and public experiences.
- This trend reflects a shift from passive attendance to active identity expression in mass events. Attendees no longer want to just consume entertainment; they want to be seen, celebrated, and empowered to express who they are. This includes gender-fluid fashion, neurodiversity-friendly environments, inclusive lineups, and culturally diverse storytelling. The individual is not just part of the crowd—they help define the crowd.

Relevance / Impact:

- High Social Relevance: Reflects broader shifts in gender politics, race discourse, and mental health advocacy.
- Policy Impact: Impacts regulations around accessibility, inclusive representation, and anti-discrimination practices at events.
- Brand Strategy: Companies and event sponsors that fail to adapt risk alienating younger audiences.
- Design Implications: Calls for rethinking accessibility, visual communication, and interactive spaces.

Weak Signals (Proofs /Examples):

- Rise of gender-neutral and body-inclusive fashion shows and runways (e.g., Chromat, Savage X Fenty). 1, 1.1
- Festivals creating neurodivergent-friendly spaces with sensory accommodations (e.g., quiet zones, low-stimulation hours). 2
- Events featuring preferred pronoun lanyards, identity workshops, and personal storytelling formats. Themed raves, parades, or art installations around intersectional identities 3



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¹ Key sources:

Purpose-Driven Participation



Key uncertainties:

- Will people stop engaging with important messages if they feel overwhelmed or pressured?
- Will purpose-driven messages work globally—or offend, clash with local norms, or even be censored?

Description:

- As social values become central to identity, people increasingly seek events that align with causes they care about—climate, justice, and community.
- Festivals and gatherings are evolving into platforms for activism, dialogue, and collective impact. Attendees expect more than entertainment—they want purpose, inclusion, and ethical alignment.



Relevance / Impact:

- Cultural Shift: Reflects the moral and emotional priorities of Millennials and Gen Z.
- Trust Building: Purpose-centered events cultivate deeper relationships between brands and communities.
- Policy Impact: Raises awareness and can directly influence public discourse and legislation.
- Market Transformation: Encourages sustainable event models, ethical consumption, and activism..

Weak Signals (Proofs /Examples):

- Music festivals, including social action zones, climate pledges (e.g., Glastonbury, SXSW). 4
- Conferences offering reparative tickets (e.g., sliding scale pricing based on privilege).5
- Eco-certifications and zero-waste commitments in event production (e.g., Greener Events accreditation).
- Partnerships with grassroots movements or Indigenous-led programming as part of the main agenda. 6

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¹ Key sources: <https://www.deloitte.com/global/en/issues/work/content/genz-millennialsurvey.html>

Community Over Crowd



Key uncertainties:

- Will people stop engaging with important messages if they feel overwhelmed or pressured?
- Will purpose-driven messages work globally—or offend, clash with local norms, or even be censored?

Description:

- "Community Over Crowd" signals a shift in how people experience mass events. Rather than simply gathering in large numbers, attendees now seek belonging, shared purpose, and personal resonance.
- What emerges is not just a crowd, but a microcosm of culture: where ideas, emotions, and values are exchanged freely.
- This shift is visible in everything from hyper-local festivals to global conferences that prioritize curated conversations, co-creation, and continuity through online platforms.
- This trend reflects a broader cultural movement away from mass broadcasting and toward niche, values-driven networks. It's not about how many attend—it's about who shows up, how they connect, and what they take away



Relevance / Impact:

- Encourages mental well-being, inclusion, and deeper human connection in an isolating world.
- Reflects the rise of micro-communities, fandoms, activist networks, and niche movements.
- Drives recurring, membership-based models over ticket-centric transactions.
- Builds loyal, self-sustaining communities where people return not just for content, but for each other.

Weak Signals (Proofs /Examples):

- Rise of micro-festivals and intentional gatherings (e.g., Campfire Convention, Envision Festival).
- Events offering co-creation roles to attendees—curating content, leading sessions, or voting on programming.
- Member-owned events and decentralized networks (e.g., DAO-based gatherings in Web3 culture).
- Use of community-building apps and platforms (e.g., Geneva, Discord, Mighty Networks) for event continuity

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¹ Key sources <https://investor.eventbrite.com/press-releases/press-releases-details/2025>

Culture fusion and Hybrid Identities



Key uncertainties:

- Will mainstream cultural institutions embrace or resist fluid identity narratives?
- Can fusion thrive in contexts where nationalism or cultural protectionism is on the rise?

Description:

- "Cultural Fusion & Hybrid Identities" reflects the rise of individuals who fluidly mix heritage, global subcultures, and lived experience into how they dress, create, perform, and gather.
- Traditional boundaries between East/West, mainstream/subculture, or sacred/secular are dissolving—replaced by dynamic, intersectional identities.
- In mass events, this is manifesting in programming that crosses genres, bilingual content, diasporic narratives, and the celebration of fluid, often contradictory cultural references.
- It's not just multiculturalism—it's cultural remixing, where identities are co-created and constantly evolving. Audiences now expect spaces where complex identities are not just accepted, but centered. Events become stages for cross-cultural performance, collective memory, and cultural regeneration—not just representation.



Relevance / Impact:

- Cultural Shift: Reflects a generational move away from fixed ethnic or national identities toward fluidity and multiplicity.
- Design Innovation: Encourages genre-blending in events—music, art, cuisine, and storytelling crossing boundaries.
- Social Cohesion: Creates shared experiences across difference, building empathy and soft power.
- Event Differentiation: Offers deeper engagement through authenticity, not mass-market flattening.

Weak Signals (Proofs /Examples):

- Music festivals booking cross-genre and cross-national collaborations (e.g., Afro-futurist DJs alongside classical orchestras).
- Hybrid fashion runways blending
- Community storytelling spaces at events for diasporic and mixed-heritage attendees to share identity journeys.
- Cultural collectives that cross borders digitally (e.g., Asia diaspora TikTok communities, transnational zine networks).

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¹ Key sources <https://theculturalpocalypse.com/globalization-and-identity-navigating-cultural-fusion-in-a-connected-world/>



Evolving Family Dynamics: Embracing Diversity

Key uncertainties:

- Diverse and non-traditional family structures are increasingly accepted, supported by evolving laws, media representation, and social norms.
- Will global policies adapt quickly to support all non-traditional families?
- How will smaller households affect future social and elder care systems?

Description:

Modern families are becoming increasingly diverse, with traditional nuclear family structures no longer being the dominant model. People are choosing different paths in terms of relationships, parenting, and household compositions. Trends like delayed marriages, single parenting, child-free couples, and the rise of same-sex families are becoming more common. This shift reflects broader societal changes, including evolving gender roles and a focus on individual choice. As a result, this transformation is significantly impacting various aspects of society, including event planning, marketing strategies, housing design, and social policies, which now need to accommodate a wider range of family structures.



Relevance / Impact:

- Cultural Shift:** As family dynamics evolve, societies become more accepting of diverse family models, leading to greater social inclusivity.
- Economic Influence:** New family structures change consumer behavior, influencing markets like housing, travel, and lifestyle products.
- Policy Adaptation:** Governments and organizations must update policies and services to support varied family units, such as healthcare, parental leave, and tax benefits.

Weak Signals (Proofs /Examples):

- In 2024, only **64% of U.S. households** were classified as family households, a significant decrease from **79% in 1974**, indicating a shift away from traditional family structures.
- **22.2% of Sweden's population** lived in single-person households, reflecting a growing trend of individuals choosing to live alone
- In several European countries, including Sweden, Norway, and France, **over 50% of children** are born outside of marriage, highlighting a move away from traditional marital norms.

Key sources: [Cultural Shift](#), [Economic Influence](#), [Policy Adaptation](#), [U.S households](#), [Sweden Population](#), [European Countries](#)

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Neo-Localism: Reclaiming the Local in a Globalized World

How Communities Are Shaping Identity, Economy, and Sustainability Through Local Action



Key uncertainties:

Will the emerging trend of neo-localism continue growing as a counterbalance to globalization, fundamentally reshaping economic structures, governance models, and cultural identities? Or will it remain a niche movement limited to certain demographics and regions?

Description:

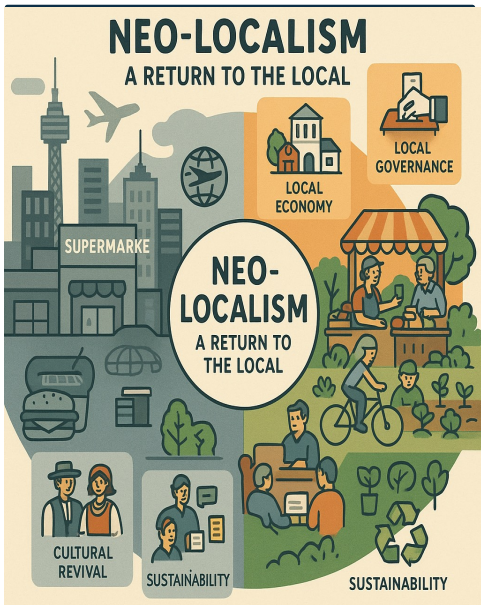
Neo-localism is a movement and philosophy emphasizing a conscious effort to promote, preserve, and deepen local identity, culture, and community attachment in response to the homogenizing effects of globalization. It seeks to reconnect people with the unique characteristics of their place—its traditions, history, products, and social bonds—often through support for local businesses, cultural events, and participatory governance.

Relevance / Impact:

- Economic Resilience: Urban parks and entertainment venues leverage neo-localism to foster community engagement and economic growth.
- Cultural Preservation: Microbreweries, harvest festivals, and local art festivals celebrate regional identity, countering cultural homogenization.
- Cross-Sector Influence: Concepts like "neo-traditional medicines" in Brazil show applications beyond economics, integrating traditional knowledge into public health.

Weak Signals (Proofs /Examples):

- Wild harvesters in the Ozarks sustaining food systems through informal practices.
- Amazon's assimilation of external innovations to drive localized business growth.
- Theatre directors adapting plays to reflect local narratives and styles.
- Tourists seeking "authentic" experiences that risk exploiting local cultures.



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Key sources: [\[1\]](#), [\[2\]](#), [\[3\]](#), [\[4\]](#), [\[5\]](#), [\[6\]](#), [\[7\]](#), [\[8\]](#), [\[9\]](#)

Urbanization : Shaping Modern Societies

The Impact of Urbanization on Cultural Identity, Social Norms, and Community Dynamics



Key uncertainties:

Population growth rates, aging populations, and migration flows (both internal and international) are difficult to predict and will shape the future demand for urban services and housing.

Description:

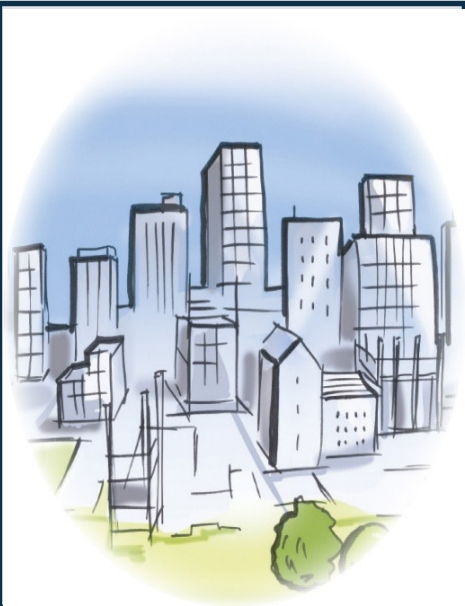
Urbanization-the increasing movement of people from rural to urban areas-remains one of the most significant trends shaping the 21st century. However, its trajectory and impacts are highly uncertain due to a range of environmental, economic, technological, political, and social factors. These uncertainties affect how cities will grow, adapt, and remain sustainable in the coming decades.

Relevance / Impact:

- **Social Inequality:** Unchecked urbanization can exacerbate poverty, inequality, and social divides, particularly if infrastructure and services fail to keep pace with population growth.
- **Public Health and Safety:** Uncertainties around disease, crime, and disaster preparedness directly impact the health and safety of urban populations.
- **Economic Competitiveness:** Cities unable to adapt to these uncertainties risk losing economic competitiveness, talent, and investment.

Weak Signals (Proofs /Examples):

- The COVID-19 pandemic saw temporary deurbanization in some regions, as remote work became feasible and people sought less dense living environments.
- The proliferation of informal settlements in rapidly growing cities like Lagos and Mumbai evidences the inability of infrastructure and services to keep pace with urban migration.



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Key sources: [\[1\]](#),[\[2\]](#),[\[3\]](#),[\[4\]](#),[\[5\]](#),[\[6\]](#),[\[7\]](#)

Technological



Key uncertainties:

- To what extent can immersive experiences truly replicate the emotional connection of in-person participation?
- How accessible will AR/VR technology be to the general public in terms of affordability, comfort, and digital literacy?
- Will data privacy and consent be ensured as virtual environments collect vast amounts of personal data?¹

Description:

Holodeck-Like Arenas: 360° Visual Environments

- 360° walls/floors covered in ultra-flexible micro-LEDs or projection-capable smart surfaces / holo projection tech
- Lidar + AI Motion tracking every person's location, gaze, and gesture

Robotic Floors & Programmable Matter:

- Programmable matter or robotics underneath to adjust physical sensations. Dynamic architecture shifts in real time—stairs rise, ground tilts, wind or vibrations simulate physical changes

Photorealistic Avatars & Holographic Presence:

- Remote users can join events through hyper-realistic VR avatars, which appear to in-person attendees as holograms or AR projections



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Relevance / Impact:

- Spaces become interactive stages where physical and digital realities merge
- Reimagines participation by allowing remote users to appear “in person”
- Reduced environmental impact by less physical travel
- New monetization models: virtual tickets, exclusive AR/VR content, digital collectibles²

Weak Signals (Proofs /Examples):

- Samsungs Transparent MICRO LED display (CES 2024)³
- Freestanding Projection domes with multi-sensory elements⁴
- Emerging light-field displays (microLED panels) that could give audiences full glasses-free 3D depth cues over large areas⁵
- Magnetically controlled metamaterial that can actively reshape itself into different forms⁶
- Disney's HoloTile Floor first-of-its-kind multi-user omnidirectional treadmill for immersive VR⁷
- Photorealistic full-body avatars from a single-camera feed⁸

Key sources: [1], [2], [3], [4], [5], [6], [7], [8]



Emotional Sync & Biofeedback: Live events become emotionally intelligent

Key uncertainties: • Will users trust systems that read and react to their emotions and biometric data? • Can emotional feedback loops be designed ethically without manipulation or overstimulation? • How will regulations evolve around neuro-rights, emotional privacy, and data ownership?	
Description: Emotion-Driven Scene Shifts (Collective Dream Mode) • Mass holodecks could shift based on crowd consensus or emotional climax • Everyone's data feeds into a central AI that morphs the scene dynamically Biometric Feedback Loops: • Advanced biometric reading (pulse, sweat, micro-expressions) lets the event itself respond to your inner state Neural Interfaces: Mind-Driven Experiences • Direct brain-computer interfaces (BCIs), like Elon Musk's Neuralink or other non-invasive versions • You feel the beat of the music as if your body is the speaker • Visuals, lights, even crowd energy are tuned to your emotional state	
Relevance / Impact: • New industries arise around sensory tech & neural interfaces • Organizers gain new behavioral insights from real-time tracking and engagement data • Redefines how we interact with media — not by clicking or choosing, but by feeling • Enables hyper-personalized, deeply immersive experiences	Weak Signals (Proofs /Examples): • Creation of personalized soundscapes by integrating real-time biometric data and using neuroscience technology¹ • OpenBCI's Galea's multi-modal sensor network & integrated software simplify collecting data from the brain & body² • Zenus Emotional AI analyzes facial expressions in real time to measure crowd sentiment, engagement, and demographics³ • Invasive BCI: Neuralink's first implant (Jan 2024): implanted patient gained thought-controlled cursor movement⁴



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- SCIENCE-FICTION

Key sources: [1], [2], [3], [4]



Smart Wearables as Personal Experience Enhancer

Key uncertainties:

How will the consent and use of personal information be governed, due to data protection regulations in Germany?

Will smart wearables be available and affordable globally?

Will it be difficult to maintain the connectivity and functionality of smart wearables in the high-density areas such as EXPO?

Description:

Smart wearables such as Brain-Computer Interfaces (BCI) and Smart Contact Lenses:

- might be able to help people with blindness and hearing loss to restore their functions.
- provide seamless real-time translation.
- provide navigation and personalized suggestions based on visitors' mood or engagement patterns.
- monitor visitors' health conditions (heart rate, hydration level, etc.), and suggest breaks or alert on-site medical staff.
- monitor and manage flow of visitors in order to avoid overcrowding.

Relevance / Impact:

- Improved accessibility and inclusivity for visitors of different background, ex. real time translation.
- Collected data can be used for the organization of the future events.
- Monitoring health conditions can help to ensure safety.
- Enhanced visitor experience, ex. personalized suggestions.

Weak Signals (Proofs /Examples):

- In 2024, Neuralink started trials of brain-computer interface (BCI) implants, with potential benefits to people with paralysis.
- Mojo Vision has patented Smart Contact Lenses with features like real-time data displays and zooming capabilities up to 14x.
- Companies like Nemaura Medical Inc, Abbott Laboratories are developing adhesive smart patches that monitor glucose levels, detect early biomarkers.



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Key sources: [1](#), [2](#), [3](#), [4](#), [5](#)

Beyond Biometrics, Toward Conscious Identity: Adaptive Neuro-BioID and the Rise of Digital Entity Authentication in a Post-Human World



Key uncertainties: "Digital Citizenship for AI Agents" – Legal identity and rights are granted to digital twins, avatars, and AI agents, enabling them to own assets, make decisions, and act autonomously under recognized BioID systems. "Neuro-Totalitarianism or Consciousness Capitalism" – In the absence of global ethical safeguards, states or corporations exploit neuro-biometric data for manipulation, surveillance, or behavior control—turning inner cognition into a new frontier of commodified identity.	
Description: Biometric identification is evolving from physical trait recognition (like fingerprints or iris scans) toward post-human, adaptive identity systems based on brainwaves, DNA, emotional states, and even consciousness. These future IDs will extend to digital twins, avatars, and AI agents, requiring novel verification methods and ethical frameworks. As identities become fluid and multilayered, technologies such as BCIs, emotion-AI, and quantum-secure IDs will play key roles in ensuring security, personalization, and ethical alignment in both physical and virtual worlds.	
Relevance / Impact: Transforms identity into a dynamic, multifaceted construct tied to behavior, cognition, and virtual presence. Enables frictionless interaction across digital and physical environments. Raises need for global ethical governance of biometric technologies.	Weak Signals (Proofs /Examples): Neurable & Kernel – Startups developing real-time brain-computer interfaces for communication and control. DNA Passport Projects – Experimental uses of genetic authentication in secure access and medicine. Worldcoin by OpenAI’s Sam Altman – An attempt to create a global identity system based on biometrics. Emotion AI in Customer Service – Companies using AI to detect emotions via voice or facial expression (e.g., Affectiva). Digital Twin Authentication in Gaming/Metaverse – Experiments with avatar-linked ID in VR platforms. Synthetic data companies like MOSTLY AI creating safe biometric datasets for model training. European AI Act & UNESCO’s Ethics of AI Framework – Early steps toward regulating biometrics and AI use.



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Key sources: [1](#) [2](#) [3](#) [4](#) [5](#) [6](#) [7](#) [8](#) [9](#)



Key uncertainties:

- The pace of AI and robotics integration into daily life may vary depending on ethical concerns, public acceptance, and regulatory frameworks.
- The potential for AI to achieve true general intelligence (AGI) remains uncertain, which could either accelerate or delay widespread adoption.
- Cybersecurity risks associated with autonomous systems may pose significant challenges to scalability.

Description:

- Artificial Intelligence (AI) and robotics involve the development of intelligent machines capable of performing tasks that typically require human intelligence, such as learning, problem-solving, and decision-making.
- By 2040, AI and robotics are expected to be deeply integrated into industries like healthcare, manufacturing, transportation, and education, with robots performing complex surgeries, managing supply chains, and even teaching.
- Advancements in machine learning, natural language processing, and sensor technology will enable robots to interact seamlessly with humans.
- AI systems will likely be capable of autonomous decision-making, with applications ranging from self-driving vehicles to smart city management.
- As AI systems grow more sophisticated, ethical considerations around transparency, accountability, and data privacy will become increasingly important.
- Meanwhile, human-AI collaboration is expected to become the norm, enhancing productivity while reshaping the nature of work.
- Education systems may also evolve to focus more on digital literacy, critical thinking, and adaptability, preparing future generations for an AI-driven world.

Relevance / Impact:

Technological: AI and robotics boost efficiency and precision, reducing human error.

Economic: Cuts labor costs but may lead to job loss in sectors like manufacturing and logistics.

Social: Improves life via personalized services, but may widen tech access gaps.

Environmental: Optimizes resources (e.g., smart grids), though AI data centers consume high energy.

Ethical: Raises concerns on privacy, algorithm bias, and the use of robots in sensitive roles.

Weak Signals (Proofs /Examples):

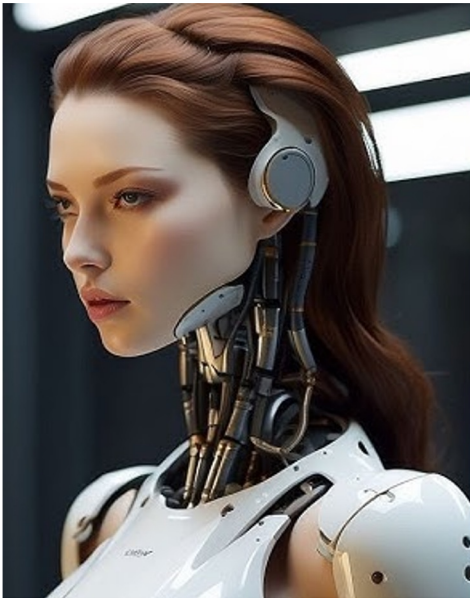
AI Models: AlphaGo¹ and GPT² showcase advanced problem-solving abilities.

Robotics: Boston Dynamics robots (Spot, Atlas³) perform tasks and navigate tough terrains autonomously.

Healthcare: AI tools like IBM Watson aid diagnosis; robotic systems like da Vinci⁴ assist in surgeries.

Autonomous Vehicles: Tesla and Waymo lead the way in AI-driven transportation.

Tesla's Optimus Gen 2⁵ and Ameca Robot⁶ are the best examples for developing future robotics technologies including AI.



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Key sources: [1](#), [2](#), [3](#), [4](#), [5](#), [6](#)

Tactile Internet

Human-Digital Interaction Through Real-Time Touch



Key uncertainties:

- Will 6G and edge computing be globally deployed in time to support real-time haptics applications? [1]
- Can global actors align on universal standards for haptic data and device interoperability? [2][3]
- How will regulators address privacy, liability, and ethical risks in remote physical interactions? [4]
- Will tactile devices become economically viable for mass adoption?

Description:

Tactile Internet is so fast and responsive that it can send and receive touch and movement in real-time, almost instantly, with no noticeable delay. Imagine you are playing a video game with someone across the world, and when you move your controller, they feel the movement instantly through a robotic arm or haptic device. Also, there could be a doctor doing remote surgery using robotic tools and they can feel the patient's body as if they were really there.

It operates using:

- Sensors that detect your touch and movements.
- Fast internet systems (ultra-low latency < 1ms) that quickly send that information to other places.
- Actuators (tiny motors or devices) that create the feeling of touch on the other side . [5][6]

Relevance / Impact:

- Transform global healthcare through remote surgeries and diagnostics. [7]
- Reinvent education and training with hands-on experiences across borders. [8]
- Support resilient economies via remote maintenance and teleoperations.
- Cross-cultural immersion through digital touch.

Weak Signals (Proofs /Examples):

- Telesurgery milestones: China's 5G-enabled telesurgery in 2019. [9]
- EU's TACTILITY project and 5G-TANGO. [10]
- Haptic Devices: HaptX, TESLASUIT, and Meta's Reality Labs. [11][12][13]

Key sources: [1] [2] [3] [4] [5] [6] [7] [8] [9] [10] [11] [12] [13]



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Key uncertainties:

- Regulation and Legal Frameworks
- Energy and Environmental Concerns
- Public Understanding and Adoption

Description:

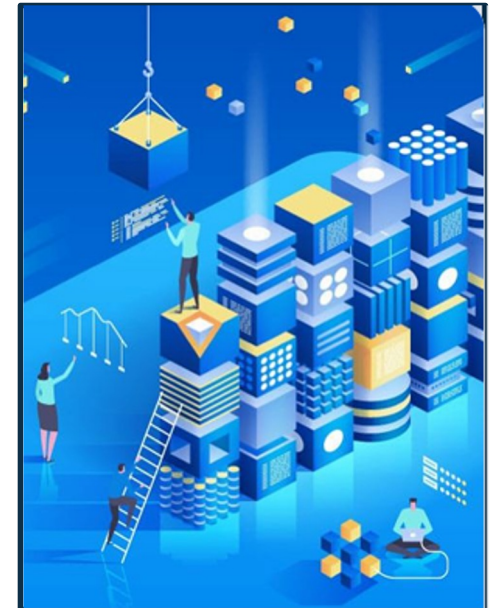
- Blockchain is not just about crypto.
- Blockchain technology has evolved significantly beyond its initial association with cryptocurrencies like Bitcoin. Its core features—decentralization, transparency, and immutability—are being harnessed across various industries to address complex challenges.
- It's a foundational tech for trusted digital systems — powering changes in finance, supply chains, identity, healthcare, and beyond.
- Its relevance grows as more industries digitize and demand faster, more secure, and more transparent operations
- In industries where trust is low (like supply chains, finance, elections), blockchain provides a verifiable record without needing a middleman.
- Removes the need for centralized authorities (banks, governments).
- This matters for digital ownership (like NFTs) and borderless financial systems (like DeFi — Decentralized Finance).

Relevance / Impact:

- **Digital:** Blockchain enables new business models: think about tokenized assets, digital identities, microtransactions, etc.
- Example: Artists selling music or art directly to fans through NFTs without platforms taking a big cut.
- Smart contracts (self-executing contracts on blockchain) automate processes, reduce paperwork, and save costs.
- **Environmental:** Blockchain verifies carbon credits and emissions in real time

Weak Signals (Proofs /Examples):

- Blockchain enables end-to-end visibility in supply chains, reducing fraud and ensuring authenticity.
- Early adopters include food safety(e.g., Walmart's blockchain for tracking leafy greens), diamond certification (e.g., Everledger), and luxury goods verification.
- **Health care:** Blockchain facilitates secure sharing of medical records, enhancing privacy and interoperability across healthcare providers.
- Projects like Medical chain and Guard time are experimenting with blockchain for health data integrity



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Key sources: <https://scet.berkeley.edu/wp-content/uploads/BlockchainPaper.pdf>
https://ntiprit.gov.in/pdf/blockchainanddistributed/Blockchain_Introduction_KR.pdf

Science Fiction

Neuro-Adaptive Learning Worlds

BCI Interfaces, Personalized Cognitive Pods, and Collective Thought Networks



Key uncertainties:

Will education evolve into fully personalized and collective cognitive ecosystems - or will access and ethics challenges deepen divides?

Description:

Education and working environments dissolve the last barriers of time, location, and even individual cognition. Learning becomes an organic, living experience: deeply personalized through DNA-based cognitive pods and expanded through global brainwave networks. AI-driven storytelling, neuro-sensory VR, and collective consciousness platforms allow people not just to study - but to live knowledge across disciplines, cultures, and realities. Workplaces and schools are no longer static spaces, but interconnected neuro-hubs, enabling real-time thought-sharing, immersive historical experiences, and deep personal growth. Learning evolves into an emotional, sensory, and collective phenomenon - driven by biological insight and group intelligence.

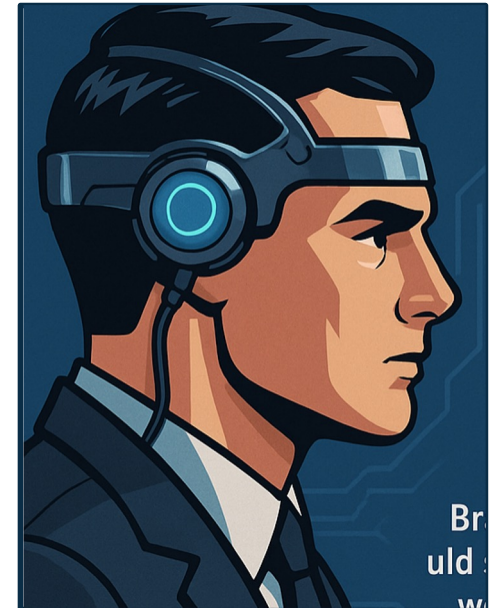
RoboCop (1987): advanced brain-computer interfaces enable workers to extend their cognitive capabilities. Inspired by RoboCop's neural integration, AI systems supplement human memory, perception, and pattern recognition - creating a symbiosis between organic and artificial intelligence in daily work environments.

Relevance / Impact:

EXPO 2035 could immerse visitors in co-created experiences of history, science, and the future. Neuro-Adaptive Worlds may make learning intuitive and emotional, promoting equity if universally accessible - or deepening inequality if not. New rules will be needed for data privacy, ethics, and emotional influence.

Weak Signals (Proofs /Examples):

- EdTech platforms using AI-driven personalization
- Wearable neurotech devices for focus and meditation
- Remote-first companies using VR for collaboration
Emotional VR experiences
- Micro-credentials and nano-degrees for flexible lifelong learning



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Key sources: RoboCop , <https://www.haufe-akademie.de/> , <https://etedge-insights.com/industry/hr/the-future-of-work-why-remote-first-companies-will-outperform-traditional-offices-in-the-next-decade/>

Neuro-Adaptive Learning Worlds

BCI Interfaces, Personalized Cognitive Pods, and Collective Thought Networks



Source: <https://www.stealthgaming.net/blog/robocop-rogue-city-a-review/>

Living Futures: Where Biology Meets Identity

Virtual Human Enhancement Labs, Fluid Avatars, and Personalized Global Futures



Key uncertainties:

Will personalized bio-enhancement and fluid identity drive global inclusion and empowerment - or create new divides based on access, ethics, and cultural fragmentation?

Description:

As physical, digital, and biological identities merge, people explore enhanced versions of themselves - physically, cognitively, and emotionally. In virtual clinics, visitors design their “Future Self” and experiment with hybrid avatars, blending cultures and realities. Identity becomes fluid and evolving, shaped by personal goals, tech, and global connection.

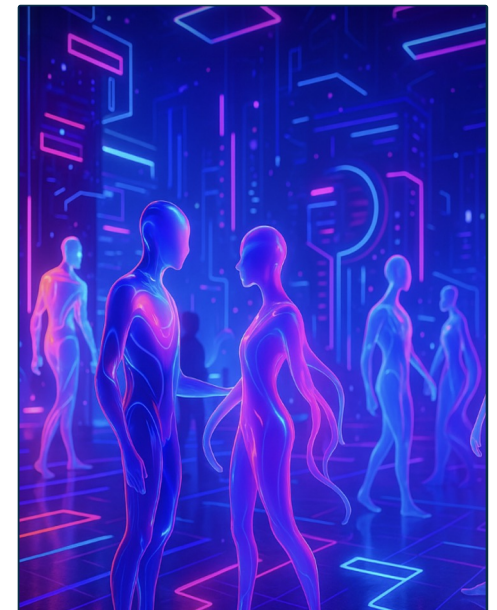
Any RPG/ Ready Player One: In hybrid virtual/physical worlds, identities are fluid. People can inhabit multiple forms, species or aesthetics - merging real-world and virtual cultures. Global cultural participation includes immersive wearables that translate virtual events into physical sensations. Smart suits allow users to 'feel' environments, interactions, and atmospheres remotely - making distant experiences like the EXPO emotionally and sensorially accessible.

Relevance / Impact:

EXPO 2035 could let visitors live their future selves. BioFusion Identity Worlds spark dialogue on diversity, ethics, and what it means to be human. Inclusive avatars and enhancement simulations may democratize opportunity - or deepen divides if access is unequal. The core question: Who do we want to become, and how do we share our future responsibly?

Weak Signals (Proofs /Examples):

- Immersive Platforms (e.g. Horizon Worlds)
- Human enhancement (e.g. Neuralink)
- Wearable tech with real-time data (e.g. Oura Rings)
- Glocalization and hybrid cultural platforms (e.g., Fortnite concerts/Minecraft global education worlds)
- Diaspora Influence (e.g., Africa No Filter, Asian diaspora media movements)



SOCIO-CULTURAL

TECHNOLOGICAL

ECONOMIC

ENVIRONMENTAL

POLITICAL /LEGAL

SCIENCE-FICTION

Key sources: Fortnite/Minecraft , <https://www.britannica.com/money/glocalization> , <https://www.uwplatt.edu/news/students-explore-glocalization-nyc> , <https://www.heise.de/en/news/15-Jahre-Minecraft-Der-ungebrochene-Erfolg-des-Kloetzchenspiels-9724035.html>

Living Futures: Where Biology Meets Identity

Virtual Human Enhancement Labs, Fluid Avatars, and Personalized Global Futures



Source:<https://medium.com/hackernoon/ready-player-one-and-the-simulation-hypothesis-virtual-worlds-avatars-cgi-b5e1b5b3dd0f>

Planetary health and circular economy

Terraforming for Regeneration



Key uncertainties:

Can regenerative robotics scale from experimental concepts to transformative forces in environmental restoration and circular economies - or remain niche and fragmented?

Description:

Sustainability is becoming a systemic approach - linking human and planetary health through circular, regenerative economies. In this vision, robots act as stewards rather than tools: regenerating soil, reforesting land, purifying water, and closing material loops with innovations like biodegradable plastics. Supported by biosensing technologies and ethical AI, these robots not only help restore Earth's ecosystems but could also apply similar principles to terraform planets. This approach is not just about minimizing harm, but about creating long-term positive impact across ecological systems.

Wall-E: A robot trying to restore the ecological balance of our planet earth.

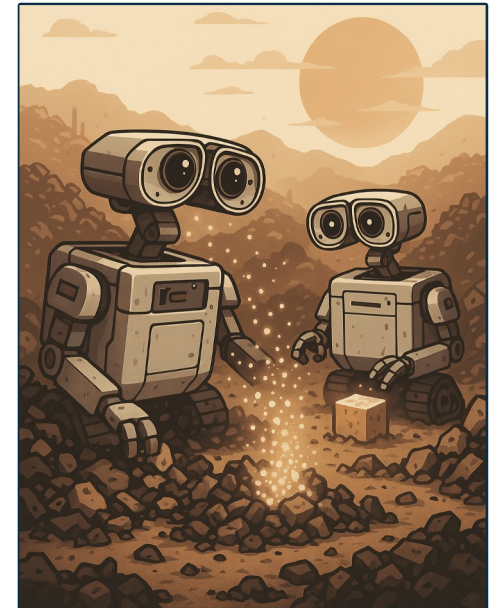
Fallout 4 (videogame): The citizens of America own a sovereign robot entity that has the ability to think and talk freely. It is built for helping humans doing basic household work, but – depending on equipment – also guard them.

Relevance / Impact:

EXPO 2035 could demonstrate zero-waste systems, regenerative food production and new lifestyles for a healthy planet. It would be possible to develop robots that counteract overconsumption and environmental pollution.

Weak Signals (Proofs /Examples):

- Circular Economy Action Plans (EU)
- Urban farming / Biodesign
- Cradle-to-Cradle building projects
- Biodegradable plastic initiatives (e.g. RIKEN)
- Miniaturized, remote-controlled nano-robots



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Key sources: Fallout 4 - Codsworth, Wall-E , <https://prezero-international.com/> , https://www.riken.jp/en/news_pubs/research_news/rr/20250327_1/index.html , <https://news.northwestern.edu/stories/2022/05/tiny-robotic-crab-is-smallest-ever-remote-controlled-walking-robot/> , <https://www.youtube.com/watch?v=Qgq2IP-k-64>

Planetary health and circular economy

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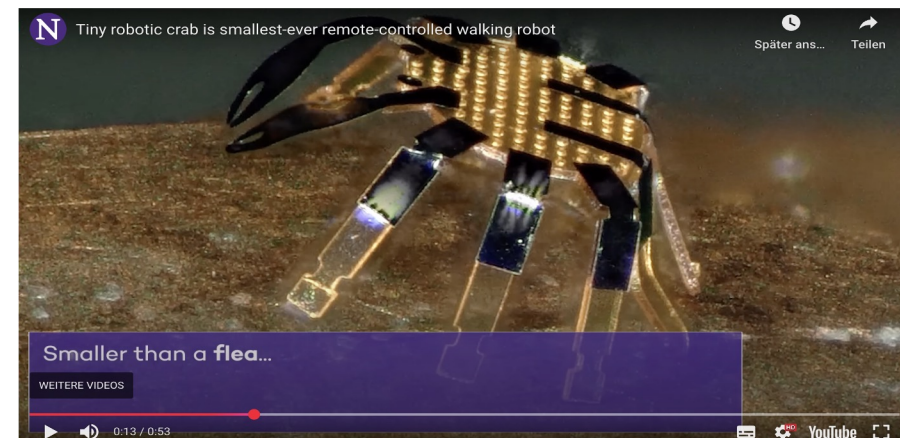
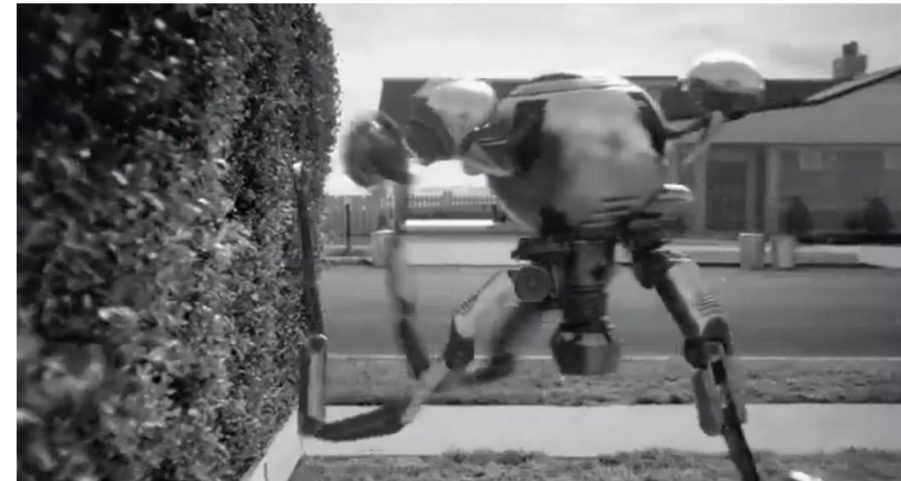


<https://www.youtube.com/watch?v=Qgq2IP-k-64> , minute: 0:21



<https://news.northwestern.edu/stories/2022/05/tiny-robotic-crab-is-smallest-ever-remote-controlled-walking-robot/>
<https://www.youtube.com/watch?v=1IP7jptXjgQ&t=13s>

https://www.youtube.com/watch?v=26_QAztlQxI,
minute 1:23



Space Agriculture

Orbital Greenhouses and Closed-Loop Farming



Key uncertainties:

Will space agriculture innovations meaningfully transform terrestrial food systems?

Description:

Space agriculture focuses on growing food in extraterrestrial environments using closed-loop systems, artificial lighting, hydroponics, and advanced monitoring. Initially developed for long-duration missions to the Moon and Mars, these technologies are now being adapted for Earth. Space-based agricultural research drives innovation in resource-efficient growing methods, supporting food security, sustainability, and resilience in extreme or urban environments. Innovations tested in orbit—such as climate control, smart sensors, and artificial lighting -are powering the rise of urban vertical farming and off-grid greenhouses. These systems can reduce dependency on global supply chains and make agriculture more resilient to climate change. In the future, space agriculture may not only support lunar or Martian colonies but also reshape how we grow food on Earth.

The Martian: An astronaut stranded on Mars uses science and ingenuity to survive, facing the isolation and harsh realities of life in space.

Relevance / Impact:

For EXPO 2035, space agriculture could show how food can be grown in hostile environments - on Mars or Earth. These innovations support climate-resilient agriculture, sustainable food systems and reduced dependency on traditional supply chains. Space agriculture can also create a new economy around off-planet biospheres and deep-tech urban farms.

Weak Signals (Proofs /Examples):

- Veggie & EDEN ISS projects: crops on the ISS/Antarctica
- NASA's MOXIE & Bioregenerative Life Support Systems: supporting closed-loop ecosystems
- SpaceX CRS missions for agricultural research to orbit
- Agri-tech startups



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Key sources: <https://www.dlr.de/en/research-and-transfer/projects-and-missions/eden-iss> , <https://science.nasa.gov/mission/veggie/> , <https://www.nasa.gov/news-release/nasa-science-cargo-launch-on-32nd-spacex-resupply-station-mission/> , <https://www.nasa.gov/missions/mars-2020-perseverance/perseverance-rover/nasas-oxygen-generating-experiment-moxie-completes-mars-mission/> , https://www.factoriesinspace.com/space-farming?utm_source=chatgpt.com

Space Agriculture

Orbital Greenhouses and Closed-Loop Farming



<https://www.scienceworld.ca/stories/can-we-farm-on-mars/>



The Data Sanctuary Dome

A safe haven for personal and collective data



Key uncertainties:

Will individuals reclaim control over their data or will monopolies strengthen? Can ethics-based data systems scale globally and remain transparent?

Description:

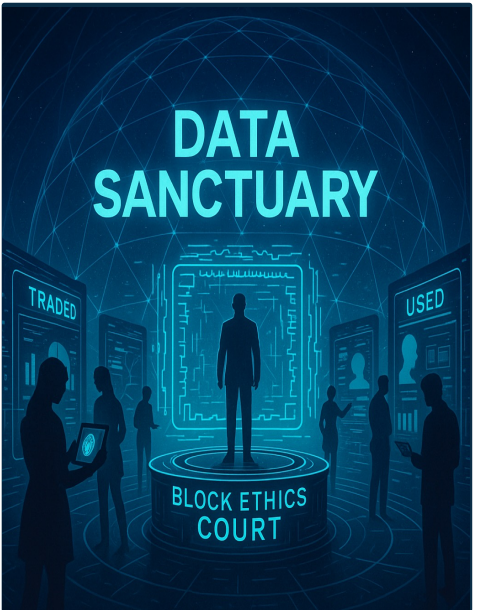
In an age of hyperconnectivity and surveillance capitalism, the Data Sanctuary Dome offers an alternative: a world where data is sacred, sovereign and self-governed. Visitors carry encrypted personal data vaults, which are secured as NFTs and accessible only via biometric keys. Visitors explore how their data is used, traded, or misused through real-time visualizations and interactive exhibits that show the consequences of different privacy choices. At the center is the Data Ethics Court, which is an AI-powered, blockchain-based tribunal where users receive verdicts on the social impact of their decisions. The dome serves as both a refuge and a simulator for ethical, transparent data futures. The Great Hack: A documentary which shows how personal data is weaponized for political manipulation, highlighting the urgent need for data sovereignty and ethical governance.

Relevance / Impact:

EXPO 2035 could be the first living lab for data dignity, where privacy becomes a shared democratic experience. Visitors leave informed and empowered, contributing to a future of ethical, decentralized data systems.

Weak Signals (Proofs /Examples):

- Personal data pods (e.g. Solid by Tim Berners-Lee)
- Data trusts (UK’s Open Data Institute)
- AI & data ethics regulation frameworks (e.g. EU AI Act, AI Bill of Rights)
- Web3 platforms focusing on self-sovereign identity (SSI)



- SOCIO-CULTURAL
- TECHNOLOGICAL
- ECONOMIC
- ENVIRONMENTAL
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- SCIENCE-FICTION

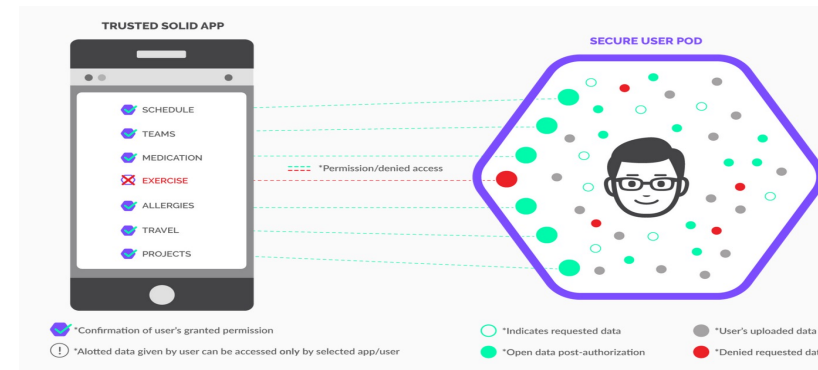
Key sources: <https://solidproject.org/>; <https://theodi.org/article/data-trusts-report/>; <https://digital-strategy.ec.europa.eu/en/policies/european-approach-artificial-intelligence>; <https://www.weforum.org/whitepapers/blockchain-beyond-the-hype>

The Data Sanctuary Dome

A safe haven for personal and collective data



The Great Hack



Personal Data Pods by Tim Berners-Lee



AI Bill Of Rights

Sources:

<https://www.imdb.com/de/title/tt4736550/> ; <https://www.privateinternetaccess.com/blog/tim-berners-lee-unveils-solid-a-decentralized-web-platform-with-privacy-built-in-as-standard/> ; <https://fi.ncsu.edu/news/blueprint-for-the-ai-bill-of-rights/>